IP Exchange Tool

Database Schema, Authentication & RBAC Documentation

Generated: October 2026 | FastAPI · SQLAlchemy 2.0 (async) · PostgreSQL · Alembic · python-jose · passlib/bcrypt

Revised edition: includes the complete Entity Relationship Model and the RBAC Dependency Factory code examples.

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1. Architecture Overview

The backend is a **FastAPI modular monolith**. Each domain lives under `app/` as a self-contained package:

```
app/
■■■ core/           # Shared: DB engine, config, validators, audit log model
■   ■■■ config.py
■   ■■■ database.py
■   ■■■ validators.py        ← NEW
■   ■■■ audit_models.py      ← NEW
■■■ auth/           # Authentication, user model, RBAC
■   ■■■ models.py            (User)
■   ■■■ refresh_token_models.py  ← NEW (RefreshToken)
■   ■■■ security.py          (hash/verify password, create JWT)
■   ■■■ service.py           (register, login, refresh, logout, change-password)
■   ■■■ dependencies.py      (get_current_user FastAPI dep)
■   ■■■ rbac.py              ← NEW (require_role, require_roles, require_verified …)
■   ■■■ schemas.py           (Pydantic I/O — enhanced)
■   ■■■ router.py            (HTTP endpoints — enhanced)
■■■ patents/        # Patent lifecycle + IP Exchange entities
```

```
■   ■■■ models.py           (Patent)
■   ■■■ exchange_models.py  ← NEW (Document, Embedding, Deal, Signature)
■   ■■■ exchange_schemas.py ← NEW (Pydantic for Exchange entities)
■■■ ipvexa/         # IPVEXA sub-system ← NEW package
■   ■■■ models.py           (IPRRecord, Service, ServiceRequest, Job, WorkReport)
■   ■■■ schemas.py
■■■ verification/   # Admin patent review workflow
■■■ search/         # (Phase 5) semantic search
■■■ ai/             # (Phase 5) AI summaries / embeddings
■■■ deals/          # (stub — Phase 4)
■■■ notifications/  # (stub — Phase 4)
```

Key design principles:

- No circular imports: FK columns reference table names (`"users.id"`), never the `User` class.
- `relationship()` uses `lazy="joined"` + `viewonly=True` for reads; writes always go through FK columns.
- Every domain exposes a `router.py` included in `app/main.py`.

---

2. Database Schema — Core (Auth)

2.1 `users`

The central identity table. All other tables reference it.

Column	Type	Constraints	Description
`id`	`INTEGER`	PK, auto-increment	Internal user ID
`email`	`VARCHAR(255)`	UNIQUE, NOT NULL, INDEX	Login identifier
`password_hash`	`VARCHAR(255)`	NOT NULL	bcrypt hash (cost 12)
`role`	`ENUM(user_role)`	NOT NULL, INDEX	`innovator` | `organisation` | `admin`
`full_name`	`VARCHAR(150)`	nullable	Display name
`phone`	`VARCHAR(20)`	nullable	E.164 format (+919876543210)
`date_of_birth`	`DATE`	nullable	Min age 18 enforced at schema level
`is_active`	`BOOLEAN`	NOT NULL, default `true`	Soft-disable account
`is_verified`	`BOOLEAN`	NOT NULL, default `false`	Email/KYC verified
`created_at`	`TIMESTAMPTZ`	NOT NULL, default `now()`	Row creation
`updated_at`	`TIMESTAMPTZ`	NOT NULL, default `now()`	Last update

Enum `user_role`**:** `innovator`, `organisation`, `admin`

File: `app/auth/models.py`

2.2 `refresh_tokens`

Persisted refresh-token store. Only SHA-256 hashes stored — raw tokens are never written to DB.

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`user_id`	`INTEGER`	FK→users.id (CASCADE), INDEX	Token owner
`token_hash`	`VARCHAR(64)`	UNIQUE, INDEX	SHA-256 hex of raw token
`expires_at`	`TIMESTAMPTZ`	NOT NULL	Expiry (default 7 days)
`is_revoked`	`BOOLEAN`	NOT NULL, default `false`	Logout / password-change revocation
`user_agent`	`VARCHAR(512)`	nullable	Browser/device hint for sessions UI
`ip_address`	`VARCHAR(45)`	nullable	IPv4 or IPv6 of login request
`created_at`	`TIMESTAMPTZ`	NOT NULL, default `now()`	

File: `app/auth/refresh_token_models.py`

2.3 `audit_logs` *(append-only)*

Security audit trail. **Never UPDATE or DELETE rows.**

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`actor_id`	`INTEGER`	FK→users.id (SET NULL), nullable, INDEX	Null for system events
`resource_type`	`VARCHAR(100)`	NOT NULL, INDEX	e.g. `"patent"`, `"deal"`
`resource_id`	`INTEGER`	nullable, INDEX	PK of acted-on entity
`action`	`VARCHAR(100)`	NOT NULL, INDEX	e.g. `"login"`, `"approve_patent"`
`outcome`	`VARCHAR(20)`	NOT NULL, default `"success"`	`success` | `failure` | `error`
`detail`	`TEXT`	nullable	JSON payload (before/after state, error)
`ip_address`	`VARCHAR(45)`	nullable	Client IP
`user_agent`	`VARCHAR(512)`	nullable	
`created_at`	`TIMESTAMPTZ`	NOT NULL, default `now()`, INDEX	

File: `app/core/audit_models.py`

3. Database Schema — Patents & IP Exchange

3.1 `patents`

Owned by: Patents track

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`owner_id`	`INTEGER`	FK→users.id, NOT NULL, INDEX	Innovator who filed
`title`	`VARCHAR(200)`	NOT NULL	
`description`	`TEXT`	NOT NULL	
`keywords`	`VARCHAR(500)`	NOT NULL, default `""`	Comma-separated tags
`status`	`ENUM(patent_status)`	NOT NULL, INDEX	Lifecycle state (see below)
`ai_summary`	`TEXT`	nullable	Generated by AI module (Phase 5)
`rejection_reason`	`TEXT`	nullable	Set on rejection
`proof_document_url`	`VARCHAR(500)`	nullable	Required before submission
`created_at`	`TIMESTAMPTZ`	NOT NULL	
`updated_at`	`TIMESTAMPTZ`	NOT NULL	

Enum `patent_status`**:** `draft` → `pending_verification` → `verified` → `listed` → `sold` / `rejected`

```
DRAFT
  ■ owner submits
  ▼
PENDING_VERIFICATION
  ■ admin approves   ■ admin rejects
  ▼                  ▼
VERIFIED           REJECTED ■■ (owner edits) ■■ PENDING_VERIFICATION
  ■ admin lists
  ▼
LISTED
  ■ deal closed
  ▼
SOLD
```

File: `app/patents/models.py`

3.2 `documents`

Legal / supporting documents uploaded by users.

Column	Type	Constraints	Description
id	INTEGER	PK	
owner_id	INTEGER	FK→users.id (CASCADE), INDEX	Uploader
patent_id	INTEGER	FK→patents.id (SET NULL), nullable, INDEX	Attached patent
deal_id	INTEGER	FK→deals.id (SET NULL), nullable, INDEX	Attached deal
doc_type	ENUM(document_type)	NOT NULL, INDEX	Type of document
file_name	VARCHAR(255)	NOT NULL	Human-readable filename
storage_url	VARCHAR(512)	NOT NULL	Object-storage URL (validated HTTPS)
mime_type	VARCHAR(100)	NOT NULL	PDF/DOCX/image — validated at upload
size_bytes	BIGINT	NOT NULL, default 0	Max 50 MB enforced in schema
created_at	TIMESTAMPTZ	NOT NULL	

Enum `document_type`**:** `patent_proof`, `assignment`, `license`, `nda`, `work_product`, `other`

Allowed MIME types: `application/pdf`, `application/msword`, `application/vnd.openxmlformats-officedocument.wordprocessingml.document`, `image/jpeg`, `image/png`, `image/webp`, `image/tiff`

File: `app/patents/exchange_models.py`

3.3 `embeddings`

pgvector embeddings for semantic patent/document search.

Column	Type	Constraints	Description
id	INTEGER	PK	
source_type	VARCHAR(50)	NOT NULL, INDEX	`"patent"` or `"document"`
source_id	INTEGER	NOT NULL, INDEX	PK of referenced entity
model_name	VARCHAR(100)	NOT NULL	e.g. `text-embedding-3-small`
vector_json	TEXT	NOT NULL	JSON-serialised float list (→ `Vector(1536)` when pgvector enabled)
created_at	TIMESTAMPTZ	NOT NULL	

> **TODO (Phase 5):** Replace `vector_json TEXT` with `vector Vector(1536)` using the `pgvector` SQLAlchemy extension and update the migration.

3.4 `deals`

Deal negotiation between an innovator (seller) and an organisation (buyer).

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`patent_id`	`INTEGER`	FK→patents.id (RESTRICT), INDEX	Patent being sold
`seller_id`	`INTEGER`	FK→users.id (RESTRICT), INDEX	Innovator
`buyer_id`	`INTEGER`	FK→users.id (RESTRICT), INDEX	Organisation
`status`	`ENUM(deal_status)`	NOT NULL, INDEX	Deal lifecycle
`price_cents`	`BIGINT`	nullable	Agreed price in smallest currency unit
`currency`	`VARCHAR(3)`	NOT NULL, default `"INR"`	ISO 4217 code
`notes`	`TEXT`	nullable	Admin/system notes
`created_at`	`TIMESTAMPTZ`	NOT NULL	
`updated_at`	`TIMESTAMPTZ`	NOT NULL	

Enum `deal_status`**:** `initiated` → `negotiating` → `pending_signature` → `signed` → `completed` / `cancelled`

Supported currencies: `INR`, `USD`, `EUR`, `GBP`, `SGD`, `AED`

3.5 `signatures`

Digital signature events for documents within a deal.

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`deal_id`	`INTEGER`	FK→deals.id (CASCADE), INDEX	
`document_id`	`INTEGER`	FK→documents.id (CASCADE), INDEX	
`signer_id`	`INTEGER`	FK→users.id (RESTRICT), INDEX	User who signed/rejected
`status`	`ENUM(signature_status)`	NOT NULL	`pending` | `signed` | `rejected`
`document_hash`	`VARCHAR(64)`	nullable	SHA-256 of the document at signing time
`rejection_reason`	`TEXT`	nullable	Set on rejection
`signed_at`	`TIMESTAMPTZ`	nullable	Timestamp of the actual signature
`created_at`	`TIMESTAMPTZ`	NOT NULL	

4. Database Schema — IPVEXA Domain

IPVEXA (IP Verification Exchange & Assistance) handles the service marketplace workflow: innovators submit IPR ideas, organisations offer services, and work is tracked through Jobs and Work Reports.

4.1 `ipr_records`

An idea or IP rights request by an innovator. Entry point for the IPVEXA workflow.

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`owner_id`	`INTEGER`	FK→users.id (CASCADE), INDEX	Innovator
`title`	`VARCHAR(300)`	NOT NULL	Min 5 chars
`description`	`TEXT`	NOT NULL	Min 20 chars
`ip_type`	`VARCHAR(100)`	NOT NULL, default `"patent"`	Free-form: patent/trademark/copyright
`status`	`ENUM(ipr_record_status)`	NOT NULL, INDEX	Lifecycle state
`admin_notes`	`TEXT`	nullable	Admin feedback
`created_at`	`TIMESTAMPTZ`	NOT NULL	
`updated_at`	`TIMESTAMPTZ`	NOT NULL	

Enum `ipr_record_status`**:** `idea` → `under_review` → `accepted` → `completed` / `rejected`

4.2 `services`

IP-related services offered by organisations.

Column	Type	Constraints	Description
`id`	`INTEGER`	PK	
`provider_id`	`INTEGER`	FK→users.id (CASCADE), INDEX	Organisation user
`name`	`VARCHAR(200)`	NOT NULL	Min 3 chars
`description`	`TEXT`	NOT NULL	
`category`	`ENUM(service_category)`	NOT NULL, INDEX	Service type
`price_cents`	`INTEGER`	NOT NULL, default 0	In paise/cents (no floats for money)
`is_active`	`BOOLEAN`	NOT NULL, default `true`	Soft-disable listing

Column	Type	Constraints	Description
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Enum service_category**:** patent_filing, trademark, copyright, trade_secret, licensing, consulting, other

---

4.3 service_requests

An innovator requesting a specific Service for their IPR Record.

Column	Type	Constraints	Description
id	INTEGER	PK	
requester_id	INTEGER	FK→users.id (CASCADE), INDEX	Innovator
ipr_record_id	INTEGER	FK→ipr_records.id (CASCADE), INDEX	Related IPR record
service_id	INTEGER	FK→services.id (RESTRICT), INDEX	Requested service
status	ENUM(service_request_status)	NOT NULL, INDEX	
message	TEXT	nullable	Max 2000 chars
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Enum service_request_status**:** pending → accepted → in_progress → completed / cancelled

---

4.4 jobs

Internal work unit created when an org accepts a ServiceRequest. **1-to-1** with service_requests.

Column	Type	Constraints	Description
id	INTEGER	PK	
service_request_id	INTEGER	FK→service_requests.id (CASCADE), UNIQUE	One job per request
assigned_to	INTEGER	FK→users.id (SET NULL), INDEX	Org user handling the job
status	ENUM(job_status)	NOT NULL, INDEX	
notes	TEXT	nullable	Internal progress log
due_date	TIMESTAMPTZ	nullable	

Column	Type	Constraints	Description
created_at	TIMESTAMPTZ	NOT NULL	
updated_at	TIMESTAMPTZ	NOT NULL	

Enum `job_status`**:** `open` → `in_progress` → `under_review` → `closed` / `cancelled`

4.5 `work_reports`

Final deliverable submitted by the org when a Job is closed.

Column	Type	Constraints	Description
id	INTEGER	PK	
job_id	INTEGER	FK→jobs.id (CASCADE), INDEX	
author_id	INTEGER	FK→users.id (SET NULL)	Org user who wrote the report
title	VARCHAR(300)	NOT NULL	
summary	TEXT	NOT NULL	
document_url	VARCHAR(512)	nullable	URL in object storage
document_mime_type	VARCHAR(100)	nullable	Validated at schema level
document_size_bytes	INTEGER	nullable	Max 50 MB
created_at	TIMESTAMPTZ	NOT NULL	

---

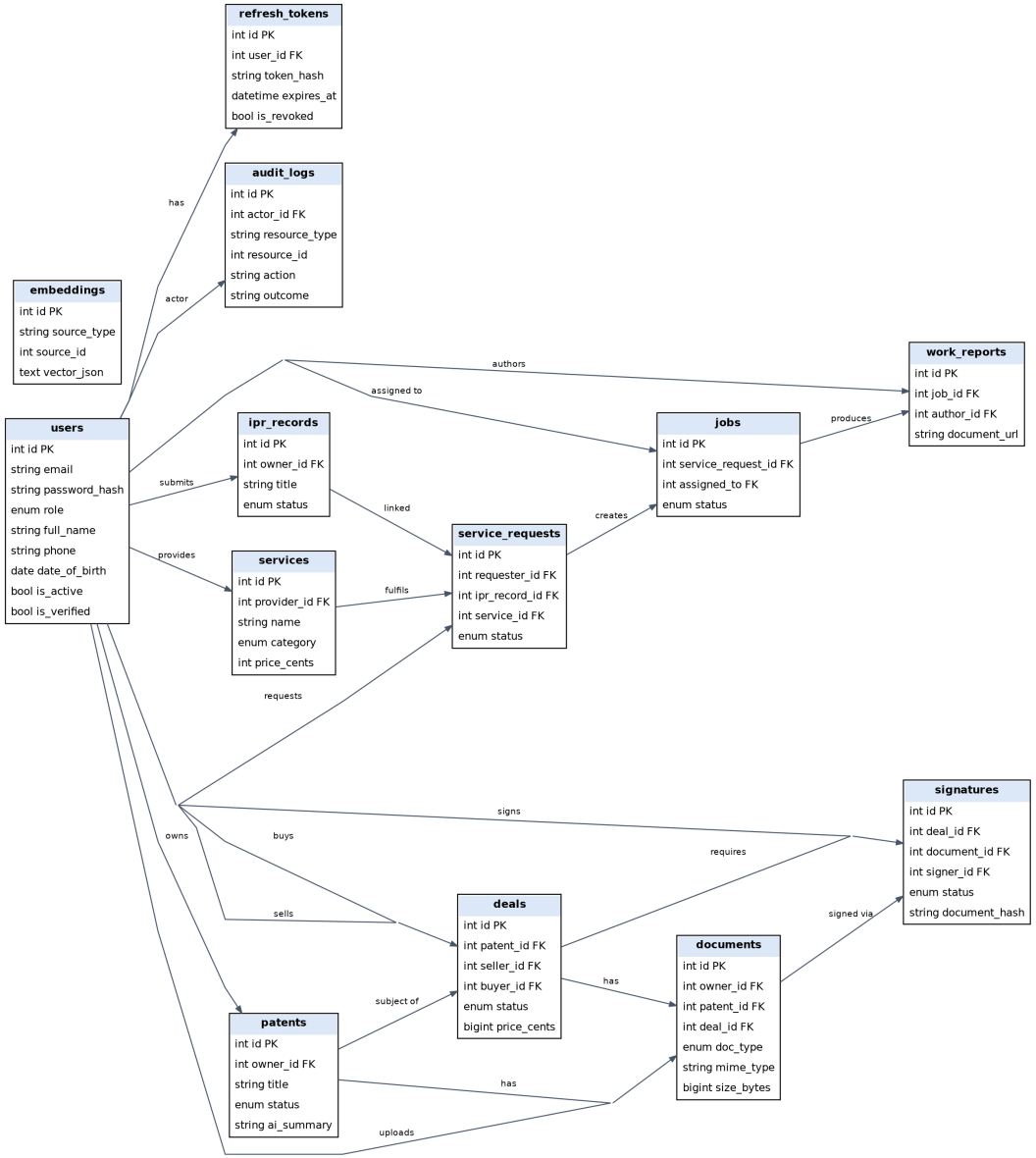
5. Database Schema — Cross-Cutting

5.1 Foreign Key Cascade Summary

Child Table	Column	Parent	On Delete
refresh_tokens	user_id	users	CASCADE
audit_logs	actor_id	users	SET NULL
patents	owner_id	users	(implicit RESTRICT)
ipr_records	owner_id	users	CASCADE
services	provider_id	users	CASCADE
service_requests	requester_id	users	CASCADE
service_requests	service_id	services	RESTRICT
jobs	service_request_id	service_requests	CASCADE
jobs	assigned_to	users	SET NULL
work_reports	job_id	jobs	CASCADE
work_reports	author_id	users	SET NULL
documents	owner_id	users	CASCADE
documents	patent_id	patents	SET NULL
documents	deal_id	deals	SET NULL
deals	patent_id	patents	RESTRICT
deals	seller_id / buyer_id	users	RESTRICT
signatures	deal_id	deals	CASCADE
signatures	document_id	documents	CASCADE
signatures	signer_id	users	RESTRICT

6. Entity Relationship Diagram

Entity Relationship Model (ERD)



Source model entities: users, refresh_tokens, audit_logs, patents, ipr_records, services, service_requests, jobs, work_reports, deals, documents, embeddings, signatures. Relationship labels are retained from the original Mermaid ER definition.

```
erDiagram
    users {
        int id PK
        string email
        string password_hash
        enum role
        string full_name
        string phone
        date date_of_birth
        bool is_active
        bool is_verified
    }

    refresh_tokens {
        int id PK
        int user_id FK
        string token_hash
        datetime expires_at
        bool is_revoked
    }

    audit_logs {
        int id PK
        int actor_id FK
        string resource_type
        int resource_id
        string action
        string outcome
    }

    patents {
        int id PK
        int owner_id FK
        string title
        enum status
        string ai_summary
    }

    ipr_records {
        int id PK
        int owner_id FK
        string title
        enum status
    }

    services {
        int id PK
        int provider_id FK
        string name
        enum category
        int price_cents
    }

    service_requests {
        int id PK
        int requester_id FK
        int ipr_record_id FK
        int service_id FK
        enum status
    }

    jobs {
```

```
    int id PK
    int service_request_id FK
    int assigned_to FK
    enum status
}

work_reports {
    int id PK
    int job_id FK
    int author_id FK
    string document_url
}

deals {
    int id PK
    int patent_id FK
    int seller_id FK
    int buyer_id FK
    enum status
    bigint price_cents
}

documents {
    int id PK
    int owner_id FK
    int patent_id FK
    int deal_id FK
    enum doc_type
    string mime_type
    bigint size_bytes
}

embeddings {
    int id PK
    string source_type
    int source_id
    text vector_json
}

signatures {
    int id PK
    int deal_id FK
    int document_id FK
    int signer_id FK
    enum status
    string document_hash
}

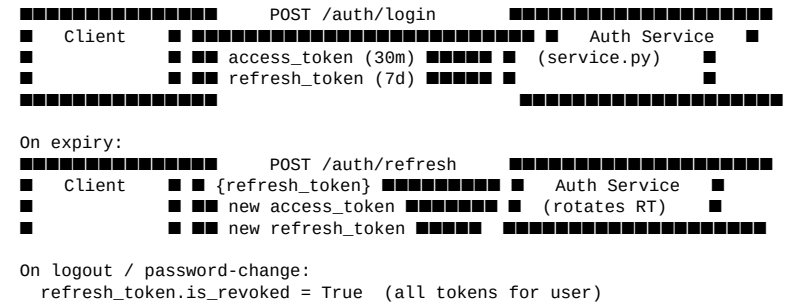
users ||--o{ refresh_tokens : "has"
users ||--o{ audit_logs : "actor"
users ||--o{ patents : "owns"
users ||--o{ ipr_records : "submits"
users ||--o{ services : "provides"
users ||--o{ service_requests : "requests"
users ||--o{ jobs : "assigned to"
users ||--o{ work_reports : "authors"
users ||--o{ deals : "sells"
users ||--o{ deals : "buys"
users ||--o{ documents : "uploads"
users ||--o{ signatures : "signs"
patents ||--o{ documents : "has"
patents ||--o{ deals : "subject of"
```

```
ipr_records ||--o{ service_requests : "linked"
services ||--o{ service_requests : "fulfils"
service_requests ||--|| jobs : "creates"
jobs ||--o{ work_reports : "produces"
deals ||--o{ documents : "has"
deals ||--o{ signatures : "requires"
documents ||--o{ signatures : "signed via"
```

---

7. Authentication — JWT + Refresh Tokens

7.1 Flow



7.2 Token Structure

Access Token (JWT)

Refresh Token

- Raw value: `secrets.token_hex(32)` (64 hex chars, 256 bits of entropy)
- Stored: SHA-256 hash only — `hashlib.sha256(raw.encode()).hexdigest()`
- Rotation: each `/auth/refresh` call revokes the used token and issues a new one

7.3 Key Files

File	Purpose
`app/auth/security.py`	`hash_password`, `verify_password`, `create_access_token`
`app/auth/service.py`	`login_user`, `register_user`, `refresh_access_token`, `logout_user`, `change_password`
`app/auth/dependencies.py`	`get_current_user` FastAPI dep (validates JWT, loads User)
`app/auth/router.py`	HTTP endpoints
`app/auth/refresh_token_models.py`	`RefreshToken` SQLAlchemy model

7.4 Endpoints

Method	Path	Auth	Description
`POST`	`/auth/register`	None	Create account
`POST`	`/auth/login`	None	Get token pair
`POST`	`/auth/refresh`	None (refresh token in body)	Rotate tokens
`POST`	`/auth/logout`	None (refresh token in body)	Revoke refresh token
`POST`	`/auth/change-password`	Bearer JWT	Change password, revoke all sessions
`GET`	`/auth/me`	Bearer JWT	Get current user profile

---

8. Role-Based Access Control (RBAC)

8.1 Roles

Role	Description	Capabilities
`innovator`	Individual IP creator	Submit patents, request IPVEXA services, sign deals
`organisation`	Corporate buyer / service provider	Browse listed patents, initiate deals, provide services, run jobs
`admin`	Platform operator	Verify/reject/list patents, manage all entities

8.2 RBAC Dependency Factory

File: `app/auth/rbac.py`

Pre-built shorthands:

8.3 RBAC Matrix

Endpoint	innovator	organisation	admin
`POST /patents`	■	■	■
`PATCH /patents/{id}`	■ (own)	■	■
`POST /patents/{id}/submit`	■ (own)	■	■
`GET /verification/queue`	■	■	■
`POST /verification/{id}/approve`	■	■	■
`POST /services` (create service)	■	■	■
`POST /service-requests`	■	■	■
`POST /jobs` (accept request)	■	■	■
`POST /deals` (initiate)	■	■	■
`POST /auth/change-password`	■	■	■

---

9. Validation Layer

File: `app/core/validators.py`

9.1 Annotated Types

Use directly as type annotations in Pydantic models:

9.2 Field Validators

Function	Input	Rule
`validate_mime_type(v)`	str	Must be in `ALLOWED_DOC_MIME_TYPES`
`validate_document_size(v)`	int	0 < v ≤ 50 MB
`validate_date_of_birth(v)`	date	Age ≥ 18 years
`validate_phone(v)`	str	E.164 format (`+[country][7-14 digits]`)
`validate_keywords(v)`	str	Max 20 keywords, each ≤ 50 chars, comma-separated
`validate_url(v)`	str	Valid HTTP/HTTPS URL
`validate_currency(v)`	str	ISO 4217 code in supported set

9.3 Constants

Constant	Value	Purpose
`ALLOWED_DOC_MIME_TYPES`	frozenset (7 types)	Whitelist for document upload
`MAX_DOCUMENT_SIZE_BYTES`	`52_428_800` (50 MB)	Maximum upload size
`MIN_AGE_YEARS`	`18`	Minimum registration age

9.4 Where Validation Is Applied

Schema	Validators Applied
`RegisterRequest`	`PasswordField`, phone E.164, DOB age check
`DocumentCreate`	`validate_mime_type`, `validate_url`, `DocumentSizeBytes`
`DealCreate` / `DealUpdate`	`validate_currency`
`IPRRecordCreate`	`validate_keywords`
`WorkReportCreate`	`validate_mime_type`, max size

---

10. Alembic Migrations

Migration History

Revision	Description	File
`3edf95aa78a4`	Initial: `users` + `patents` tables	`3edf95aa78a4_create_users_and_patents_tables.py`
`a1b2c3d4e5f6`	IPVEXA, Exchange, Audit, Refresh Token tables + user profile columns	`a1b2c3d4e5f6_add_ipvexa_exchange_audit_and_refresh_token_tables.py`

Running Migrations

Migration `a1b2c3d4e5f6` — Tables Created

1. `users` → +`full_name`, `phone`, `date_of_birth` columns
2. `refresh_tokens`
3. `audit_logs`
4. `ipr_records`
5. `services`

```
6  service_requests
7  jobs
8  work_reports
9  deals
10 documents
11 embeddings
12 signatures
```

---

11. Environment Variables

File: `.env.example`

Variable	Default	Required	Description
`DATABASE_URL`	—	■	`postgresql+asyncpg://user:pass@host/db`
`ENVIRONMENT`	`development`	—	`development` | `production`
`DEBUG`	`true`	—	SQLAlchemy echo, debug mode
`JWT_SECRET_KEY`	placeholder	■ (prod)	Long random string (≥ 32 chars)
`JWT_ALGORITHM`	`HS256`	—	JWT signing algorithm
`ACCESS_TOKEN_EXPIRE_MINUTES`	`30`	—	Access token TTL in minutes
`REFRESH_TOKEN_EXPIRE_DAYS`	`7`	—	Refresh token TTL in days
`COOKIE_SECURE`	`false`	—	Set `true` in production (HTTPS)
`OPENAI_API_KEY`	—	Phase 5	
`ANTHROPIC_API_KEY`	—	Phase 5	

---

12. Module Ownership Map

Package	Files	Owner
`app/core/`	`config.py`, `database.py`, `validators.py`, `audit_models.py`	Core / Shared
`app/auth/`	All files	Auth track
`app/patents/`	`models.py`, `router.py`, `schemas.py`	Patents track
`app/patents/exchange_models.py`	Document, Embedding, Deal, Signature models	IP Exchange track
`app/patents/exchange_schemas.py`	Pydantic schemas for exchange entities	IP Exchange track
`app/ipvexa/`	All files	IPVEXA track
`app/verification/`	All files	Verification track
`app/search/`	(Phase 5)	Search track
`app/ai/`	(Phase 5)	AI track
`app/deals/`	(Phase 4 stub)	Deals track
`app/notifications/`	(Phase 4 stub)	Notifications track

Rule: If you are not the owner of a package, open an issue rather than editing its files directly. `app/main.py` and `alembic/` are shared — coordinate before touching them.

Appendix A — Quick Reference: Import Paths